

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6/1	(a)	(i)		$2500 \div 15$ (1) 166.7 months (1) $\div 12 = 13.89$ yrs (1) Allow 13.8888888 / 13.9 / 14 / 13 yr 11 months OR 15×12 (1) = 180 $2500 \div 180$ (1) = 13.89 (1) Accept 13.8 / 13.88 / 13 / 166.7 for (2) Accept 166.6 / 180 for (1)		3		3	3	
		(ii)		$3.6 \div 4$ (1) (substituting selected data) $\times 100 = 90\%$ (1) Accept 0.9 for (1)		2		2	2	
		(iii)		$2 \times 3 = 6$ (1) 6 (ecf from multiplication shown) $\times 22$ OR 6×0.22 (1) = 132p / $\text{£}1.32$ (1) (ans and correct unit) Accept for 2 marks : $\text{£}0.66$ / 66p / 132 (no unit) / 1.32 (no unit) / 105.6 p / $\text{£}1.056$ / 106 p / $\text{£}1.06$ Alternate for 2 marks: $3 \times 1.6 = 4.8$ (0) 4.8 (ecf) $\times 22$ OR 4.8×0.22 (1) = 105.6 p / $\text{£}1.056$ (1) Accept for 1 mark 105 p / $\text{£}1.05$		3		3	3	

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		(iv)		Any 2 × (1) from: Heater 1 uses more energy / heater 1 is less efficient / heater 2 uses less energy / heater 2 is more efficient (1) Heater 1 <u>heats</u> up more water / heater 2 <u>heats</u> up less water / heater 1 <u>heats</u> up 150 and heater 2 <u>heats</u> up 50 litres / heater 2 only <u>heats</u> up 50 litres / 50 litres is closer to 40 litres (1) Heater 1 wastes more hot water / heater 2 wastes less hot water / heater 1 wastes 110 litres of water and heater 2 wastes 10 litres of hot water / heater 2 only wastes 10 litres of hot water (1) Heater 1 costs more money / Heater 2 costs less money / Heater 1 costs 132p and heater 2 costs 44p / heater 2 only costs 44p (1) Heater 1 takes more time / heater 2 takes less time / heater 1 takes 3 hours to heat the water and heater 2 takes half an hour / heater 2 only takes half an hour to heat the water (1) (So disagree) references to payback time neutral			2	2		
	(b)			Furred up / blocked pipes in boiler /rust /decreased water flow/ less efficient / takes longer to heat up the water / harder to heat up the water / hard water has a higher boiling point (1) (due to build up of) limescale / calcium or magnesium carbonates / scum / calcium and magnesium ions (1) Neutral: causes damage / makes bill go up	2			2		

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	(c)	(i)		Foam is a good insulator/poor conductor / reduces conduction (1) Not: stops conduction Because it has air trapped in in it (1) OR Shiny surfaces are poor emitters (1) of IR / radiation (1) OR Foam is a good insulator/poor conductor (1) Shiny surfaces are poor emitters (1) OR Shiny surfaces reflect (1) {IR / radiation} back in (1)	2			2		
		(ii)		Uses less electricity / less energy / less power (1) Less fuel required – fewer power stations / less mining (of fuel) / less pollution / less (additions to the) greenhouse effect (1)		2		2		
		(iii)		Convection (1) (Only) heats water above the heater / can't heat water below the heater (1) Because hot water rises / is less dense (1) not: heat rises or heat is less dense (So agree)			3	3		
				Question 6 total	4	10	5	19	8	